

Hardened Audit and Reproducible Acumen/SSI Oracle Test Protocol

Process hardened-audit-v9-repository-gate-and-oracle-runner-hardening

Current repository head reviewed: dcacbf4458f2049aeec01be345b32d1685dae27c

Last documented green code head: 2dc369678dfc294db189d1bc706eba4ab02b752a

Executive conclusion

The hardened process passed 22 of 22 independent integrity checks. The project-library audit covers 436 extracted paths, 175 unique SHA-256 payloads, 40 native MPP files, 8 schedule XML files, 83 XLSX workbooks, 6 Acumen AFW workspaces, and all remaining reference formats.

The process is internally verified, but the corpus is not fully readiness-green. Four SSI exports lack exact dated source snapshots; one AFW cannot identify one binary Project5_TAMPERED source; the current repository head was not locally executed; and the current updated4 24-hour MPP has not been semantically compared to the archived updated3 payload. These are hard blockers.

Measure	Result
Extracted paths / unique payloads	436 / 175
Schedules	40 MPP + 8 MSPDI schedule XML
Vendor outputs	56 Acumen XLSX + 19 SSI XLSX + 6 AFW
Independent checks	22/22 passed
Exact SSI source readiness	15/19
Exact AFW binary readiness	5/6
Current repository full gate	Not executed in this runtime

Weaknesses found and hardened controls

#	Weakness	Control	Verification
1	Filename/folder inference can create false associations.	Associations require embedded project/version labels, exact UID/name/date reproduction, or explicit AFW serialized file references. Folder/name similarity alone is prohibited.	No association has source_resolution=heuristic_only; all 81 vendor-oracle containers are evidence-linked.
2	A same filename can represent several distinct binary schedules.	Use SHA-256 as binary identity and record schedule-equivalence classes separately from binary identity.	11 source-equivalence classes are explicit; AFW exact-binary gate correctly fails 5/6.
3	A re-saved MPP can change bytes without changing schedule semantics, or change semantics silently.	Compare semantic task, logic, calendar, resource, assignment, property, and combined fingerprints; never treat a Git rename as semantic equivalence.	updated4 24-hour file remains UNVERIFIED because current binary was unavailable; no equivalence claim is made.
4	Spreadsheet filenames and visible report titles can be stale or misleading.	Read every populated cell, formulas, workbook relationships, core properties, project strings, date strings, sheet structures, and vendor-specific rows.	83/83 XLSX scanned; 3,909,788 cells and formula digests recorded.
5	A worksheet can contain multiple project versions.	Model one oracle container to many schedule associations and retain version-specific labels/status dates.	Acumen multi-project reports create multiple evidence edges rather than one guessed source.
6	SSI rows can match UIDs/names but not scheduled dates.	Require independent Start and Finish date reproduction for exact source status; classify lower matches as probable or structural-only.	15/19 SSI workbooks pass >=99.5% exact date match; four are explicitly blocked from exact parity.

#	Weakness	Control	Verification
7	MPP-to-MSPDI XML exports are lossy.	Use XML only for logic/task portability tests; retain MPP for calendars, resources, assignments, stored critical/slack, and project properties.	13/13 same-stem pairs have exact logic fingerprints, while field-loss differences remain documented.
8	Acumen display name alone is not a metric identity.	Key metrics by library/group/GUID/version and preserve Primary, Secondary, Tripwire, filters, thresholds, columns, and detailed-report options independently.	AFT architecture and Missing Logic variants are represented in the formula audit and protocol.
9	Headline equality can hide offender-list or denominator mismatches.	Oracle contract compares value, count, population, status, offender UID set, row order where relevant, and threshold band separately.	Test-case schema requires granular expected artifacts, not only headline values.
10	Floating-point and date-format differences can create unstable comparisons.	Canonicalize datetimes, use integer working minutes where possible, Decimal/ROUND_HALF_UP at presentation, and explicit field-specific tolerances.	Comparison contract in V9 records exact/tolerant fields and N/A semantics.
11	A single parser implementation can validate its own error.	Independent verification uses raw ZIP/XML counts, SHA-256/CRC, MPXJ semantic fingerprints, and cross-tool oracle evidence.	HARDENED_VERIFICATION_RESULT_V9.json records independent controls.
12	Cached analysis can conceal source changes.	Cache keys include file SHA-256 and engine version; hardened audit independently recomputes source hashes before accepting cached metadata.	All canonical and alias hashes are rechecked by verify_harden_v9.py.
13	Repository documentation may describe a prior code head while reference uploads advance main.	Track code-audited head separately from current repository head; rerun the full gate after every reference-only upload.	Current reference-upload head has no connector-visible workflow run and is marked unexecuted.
14	The repository cannot be exhaustively executed from an environment without a checkout/network.	Ship a repo-local audit script and matrix runner that Claude Code executes inside the repository; external review is not represented as a local full-gate run.	Repository execution gate remains failed until the generated script is run in the repo.
15	Vendor stochastic outputs may not be bit-exact because RNG/seed/correlation implementation is proprietary.	Separate deterministic formula/OAT checks from distributional SRA checks; require fixed seed/config and statistical acceptance bands for Monte Carlo.	Protocol distinguishes exact deterministic parity from statistical distribution parity.
16	The original combined semantic fingerprint included MPXJ ProjectProperties.CurrentDate, a read-time value that changed on every parse and made all 48 schedules look non-repeatable.	Exclude CurrentDate from the normative stable_combined_v2 fingerprint. Keep the old combined digest only as a diagnostic; gate on counts, stable properties, and identity/logic/schedule/calendar/resource/stable-combined fingerprints.	All 48 schedule inputs were reparsed independently. Counts and five component fingerprints matched 48/48; the only original combined difference was CurrentDate. stable_combined_v2 matched 48/48.

#	Weakness	Control	Verification
17	The original formula_count counted only non-empty formula definitions, while shared-formula follower cells contain empty <f> elements. Two workbooks therefore appeared to have inconsistent formula counts under a raw OOXML recount.	Track formula_definition_count and formula_cell_count separately. Validate both: definitions preserve unique formula text, while formula-bearing cells measure calculation coverage.	All 83 XLSX files independently reproduce cell count, sheet count, formula definition count, and formula-bearing cell count. Margin template has 202 definitions/350 formula cells; PerformanceAnalysisSummary has 214 definitions/7,244 formula cells.
18	File-level vendor classification and refined workbook-level classification diverged for PerformanceAnalysisSummary_Sample Metrics_2026APR.xlsx.	Use one authoritative classification table and propagate it to file, workbook, test-case, and association layers; direct vendor oracles are separated from templates/reference workbooks.	PerformanceAnalysisSummary is now Reference/Methodology / Performance Analysis Template and is excluded from the Acumen-oracle readiness denominator.
19	The first repo-local audit script did not exactly reproduce the complete documented CI gate and did not distinguish tracked, untracked, dirty, or missing worktree files.	Record Git head/branch/status/tracked files, fail dirty or incomplete worktrees, structurally parse all files, and run the documented coverage, parity, security, dependency-audit, JavaScript syntax, and native-MPP test commands.	verify_hardened_v9.py statically confirms the full gate and Git-state controls are present in repo_hardened_audit.py; execution remains a readiness blocker until run in a current checkout.
20	The first oracle runner could compare headline JSON without source-hash provenance and compared SSI dates only as raw strings, allowing the wrong source snapshot or equivalent date encoding to pass incorrectly.	Require an allowed source SHA-256 for exact SSI cases, normalize Excel serial and ISO datetimes to one numeric axis, normalize day quantities, detect duplicate UIDs, optionally enforce row order/Drag, and support strict metric provenance.	verify_hardened_v9.py executes exact metric and SSI self-tests through oracle_matrix_runner.py and requires zero differences.
21	Versioned control text could remain stale after a later hardening pass, as shown by V6 artifact names inside the V8 process-control table.	Treat process-control artifact references as versioned data and fail verification when obsolete V6/V7/V8 process-artifact names remain in the V9 control set.	The V9 verifier scans process_weaknesses_and_controls for obsolete versioned verifier/result references.

Independent verification results

Check	Expected	Actual	Result
Canonical files exist	175	175	PASS
Canonical SHA-256 reproducible	175	175	PASS
Canonical byte sizes reproducible	175	175	PASS
Alias paths reproduce canonical SHA-256	436	436	PASS
Source ZIP CRC/traversal clean	0	0	PASS
XLSX raw cell/formula/sheet counts independently reproduce	83	83	PASS
All 48 schedule stable semantic fingerprints independently reproduce	48	48	PASS

Check	Expected	Actual	Result
Volatile CurrentDate difference identified and excluded from normative gate	48	48	PASS
Non-schedule XML configurations well-formed	2	2	PASS
AFW explicit references independently reproduce	6	6	PASS
Association IDs valid	169	169	PASS
All vendor oracle containers associated	81	81	PASS
No heuristic-only associations	0	0	PASS
Exactly one test contract per vendor oracle	81	81	PASS
Test contracts pin oracle SHA-256	89	89	PASS
MPP/XML pairs retain exact logic fingerprints	13	13	PASS
SSI exact-source readiness independently equals 15/19	15	15	PASS
AFW exact-binary readiness independently equals 5/6	5	5	PASS
Readiness gate arithmetic coherent	6	6	PASS
Process-control artifact references are current V9	0	0	PASS
Repo-local audit script contains full Git/coverage/parity/security/JS/MPP controls	9	9	PASS
Oracle runner validates schema and passes exact metric/SSI self-tests	3	3	PASS

Readiness gates

Gate	Expected	Actual	Result	Reason
Every SSI oracle has a >=99.5% exact start/finish source match	19	15	FAIL	
Every Acumen XLSX has exact embedded project/version labels	56	56	PASS	
Repository current head locally checked out and full test gate rerun	1	0	FAIL	Runtime DNS blocked git clone; GitHub connector evidence reviewed but current reference-upload head was not executed locally.
Every AFW explicit source filename resolves to at least one schedule equivalence class	6	6	PASS	All six AFW workspaces expose explicit .mpp references that resolve to one or more exact-name candidates.

Gate	Expected	Actual	Result	Reason
Every AFW explicit source filename resolves to exactly one binary payload	6	5	FAIL	ref2/2345.afw references Project5_TAMPERED.mpp, but three distinct archive payloads share that filename and reproduce the same observable schedule rows. The AFW stores no source content hash.
Current repository updated4 24-hour file semantically compared to archived updated3 24-hour file	1	0	FAIL	Git blobs differ and current repository binary was not downloadable through the available connector.

Full project-library corpus

- 436 paths and 175 unique payloads; all canonical and alias hashes reverified.
- 40 MPP and 8 schedule XML independently reparsed. Two field-map XML files are valid configuration files.
- 83 XLSX workbooks and 3,909,788 cells verified, including separate formula-definition and formula-bearing-cell counts.
- 56 Acumen XLSX, 19 SSI XLSX and 6 AFW workspaces cross-referenced to schedule evidence.
- No XER exists in the uploaded project library; production P6 parity remains an evidence gap.

Newly corrected audit defects

Volatile CurrentDate fingerprint

The original combined schedule fingerprint included MPXJ ProjectProperties.CurrentDate. That field is generated at read time and changed on all 48 independent reparses. The stable component fingerprints were exact. stable_combined_v2 now excludes CurrentDate and matched 48/48.

Shared-formula counts

formula_count originally meant non-empty formula definitions, not every formula-bearing cell. Shared-formula follower cells use empty <f/> elements. The hardened inventory stores formula_definition_count and formula_cell_count separately; all 83 workbooks independently match both.

Template misclassification

PerformanceAnalysisSummary_Sample Metrics_2026APR.xlsx contained Acumen-derived content but is a reference/template, not an Acumen-generated oracle. File-level and workbook-level classification are now synchronized.

Oracle-to-source association model

Allowed evidence: Acumen embedded project/version labels and status dates; SSI UID/name/Start/Finish reproduction; explicit AFW MPP references; and documented semantic-equivalence classes. Filename/folder similarity, project title alone, row number, and task name alone are not sufficient.

SSI exactness and gaps

SSI workbook	Best candidate schedule	Rows	UID	Name	Dates	Ratio	Status
ref2/UID_152_Directional_Path_Analysis_All Dependents Not Leveled_2026-6-23-20-30-55.xlsx	ref1/Large Test File Leveled with Driving Slack.mpp	783	783	783	757	96.68%	probable
ref2/UID_152_Directional_Path_Analysis_Leveled_All Dependents_2026-6-23-20-28-26.xlsx	ref1/.mpp test files/Large Test File/Large Test File Leveled.mpp	783	783	783	753	96.17%	probable
ref2/UID_4_Directional_Path_Analysis_2026-6-23-15-42-46.xlsx	ref1/5 Task Test File.mpp	4	4	4	1	25.00%	structural-only
ref2/UID_4_Directional_Path_Analysis_Progress_2026-6-23-16-0-6.xlsx	ref1/5 Task Test File.mpp	4	4	4	1	25.00%	structural-only

These four are structural/probable tests only. Restore the exact source snapshots or regenerate the exports before exact parity use.

AFW and naming ambiguities

ref2/2345.afw resolves Project5_TAMPERED.mpp to three distinct archive payloads. It proves an equivalence class but not a unique binary. The Large Test File OverAl Fixed SSI workbook also exactly matches a differently named Not Leveled source, proving filename inference unsafe.

Repository audit boundary

The current repository was reviewed through GitHub and the repository-owned full inventory, with direct checks of build configuration, CI, state documentation, formula-audit code, parity tests and golden cases. The code architecture is importer -> immutable Schedule model -> CPM/metrics -> FastAPI web layer, with local-only AI and strict parity gates.

GitHub commit comparison shows the current head is 11 commits ahead of the last documented green code head and changes only nine reference paths: seven XER removals, one modified Hard_File_updated3.mpp, and the 24-hour MPP rename. No source, test, build, CI, or documentation code path changed. The code state therefore equals the last documented green code head, which reports 2,208 tests and all gates green. The current binary reference changes still were not locally executed because the runtime could not obtain a checkout and no workflow run was visible. Run repo_hardened_audit.py in Claude Code to close that final gate.

Acumen metric identity and comparison

Key by group + GUID + version + display name. Preserve Primary Formula, Secondary Formula, Tripwire/Highlight Formula, separate filters/inclusions, phase/date basis, prorating, thresholds, columns and detailed-report options. Compare count/numerator, population/denominator, raw value, displayed value, offender UID set, threshold band, Fail flag and detailed rows independently.

The repository pins the 20260423 AFT as the normative Bible. The 20260708 AFT is later reference evidence and requires an ADR plus regenerated parity before promotion.

SSI comparison contract

Persist SSI version, source hash, focus UID, direction, all-dependencies mode, constraint and leveling-delay handling, tier bands, calendar convention and export timestamp. Exact tests compare UID set, task name, Start, Finish, Driving Slack, Trace Log tier and ordered driving-path sequence.

Predictable outputs

Case	Mode	Expected summary	Source
ACUMEN-PROJECT2	exact	<pre>{"schedule_quality": {"missing_logic": 6, "logic_density": 2.79, "critical": 41, "hard_constraints": 0, "negative_float": 0, "insufficient_detail": 1, "number_of_lags": 2, "number_of_leads": 0, "merge_hotspot": 10}, "dcma14": {"DCMA01": 4, "DCMA02": 0, "DCMA03": 2, "DCMA04_FS_pct": 99, "DCMA04_SSFF": 1, "DCMA04_SF": 0, "DCMA05": 0, "DCMA06": 44, "DCMA07": 0, "DCMA08": 1, "DCMA09": 0, "DCMA10": 0, "DCMA11": 18, "DCMA12": "PASS", "DCMA13": 1.0, "DCMA14": 0.74}, "baseline_compliance": {"forecast_to_be_finished": 27, "completed_on_time": 9, "completed_late": 11, "not_completed": 7, "baseline_finish_compliance_p ct": 33, "forecast_to_be_started": 29, "started_on_time": 11, "started_late": 12, "not_started": 6, "baseline_start_compliance_pc t": 41}}</pre>	tests/fixtures/golden/ project2_5/case.json

Case	Mode	Expected summary	Source
ACUMEN-PROJECT5	exact_with_documented_basis_variants	{ "schedule_quality": { "missing_logic": 7, "logic_density": 2.81, "critical": 4, "hard_constraints": 1, "negative_float": 0, "insufficient_detail": 0, "number_of_lags": 2, "number_of_leads": 0, "merge_hotspot": 10, "dcma14": { "DCMA01": 5, "DCMA02": 0, "DCMA03": 1, "DCMA04_FS_pct": 99, "DCMA04_SSFF": 0, "DCMA04_SF": 0, "DCMA05": 1, "DCMA06": 44, "DCMA07": 0, "DCMA08": 0, "DCMA09": 0, "DCMA10": 0, "DCMA11": 37, "DCMA12": "PASS", "DCMA13": 1.0, "DCMA14": 0.59, "baseline_compliance": { "forecast_to_be_finished": 46, "completed_on_time": 9, "completed_late": 18, "not_completed": 19, "baseline_finish_compliance_pct": 20, "forecast_to_be_started": 48, "started_on_time": 11, "started_late": 18, "not_started": 19, "baseline_start_compliance_pct": 25, "change_from_project2": { "activities_added": 0, "new_critical": 1, "no_longer_critical": 34, "finish_date_slips": 9, "start_date_slips": 9, "remaining_duration_increases": 9, "float_erosion": 1, "completed": 27, "in_progress": 2, "net_finish_impact_engine_cpm_days": -148, "net_finish_impact_fuse_stored_days": -134 } } } } }	tests/fixtures/golden/project2_5/case.json
ACUMEN-HARD-FILE	exact_with_documented_divergences	{ "status_date": "2026-07-07", "missing_logic": 7, "hard_constraints": 0, "high_float_44d": 2, "milestones_with_duration_gt_0": 0, "tasks_and_milestones_to_go": 110, "milestones_to_go": 25, "normal_tasks_to_go": 85, "normal_tasks_to_go_in_progress": 0 }	tests/fixtures/golden/fuse_hardfile/case.json

Case	Mode	Expected summary	Source
ACUMEN-HARD-FILE-UPDATED	exact_with_documented_divergences	{"status_date": "2026-08-11", "hard_constraints": 0, "high_float_44d": 6, "milestones_with_duration_gt_0": 0, "tasks_and_milestones_to_go": 103, "milestones_to_go": 24, "normal_tasks_to_go": 79, "normal_tasks_to_go_in_progress": 1, "bei_value_tasks": 0.27, "spi_t_acumen": 0.8}	tests/fixtures/golden/fuse_hardfile/case.json
ACUMEN-HARD-FILE-UPDATED2	exact_uid_sets	{"status_date": "2026-09-10", "bei_value_tasks": 0.59, "spi_t_acumen": 1.14, "invalid_forecast_dates_count": 21, "critical_path_count": 54, "negative_float_count": 54, "fuse_missing_logic_count": 6, "engine_missing_logic_count": 9, "documented_extra_uids": [187, 400, 412]}	tests/fixtures/golden/fuse_hardfile/case.json
ACUMEN-HARD-FILE-UPDATED3	exact_uid_sets	{"status_date": "2026-10-12", "bei_value_tasks": 0.47, "spi_t_acumen": 1.25, "invalid_forecast_dates_count": 0, "critical_path_count": 50, "negative_float_count": 50, "fuse_missing_logic_count": 6, "engine_missing_logic_count": 9, "documented_extra_uids": [187, 400, 412]}	tests/fixtures/golden/fuse_hardfile/case.json
ACUMEN-EVM1	exact	{"status_date": "2012-09-01T17:00", "non_summary_tasks": 11, "critical": 10, "DCMA01": 2, "DCMA05": 0, "DCMA07": 0, "DCMA06": 0, "DCMA04_FS_pct": 100, "BEI": 0.0, "project_finish": "2012-09-12"}	tests/engine/test_evm_acumen_reference.py
ACUMEN-EVM2	exact_plus_known_progress_scheduler_residual	{"status_date": "2012-09-12T17:00", "non_summary_tasks": 11, "critical": 8, "DCMA01": 1, "DCMA05": 0, "DCMA07": 0, "DCMA06": 0, "DCMA04_FS_pct": 100, "BEI": 0.25, "engine_project_finish": "2012-10-01", "acumen_project_finish": "2012-10-04", "engine_net_finish_impact": -19.0, "acumen_net_finish_impact": -22.0, "acumen_spi_t_target": 0.56}	tests/engine/test_evm_acumen_reference.py

Case	Mode	Expected summary	Source
SSI-CORPUS	row_exact_or_explicitly_blocked	{ "test_case_file": "HARDENED_TEST_CASES_V9.json", "direct_ssi_oracles": 19, "exact_source_ready": 15, "blocked_or_probable": 4, "comparison_key": "Unique ID", "exact_fields": ["Task Name", "Start", "Finish", "Driving Slack", "Trace Log Value"], "ordered_field": "driving path sequence when exported as an ordered trace"} }	uploaded project-library SSI workbooks

Execution runbook

Step	Action
1	Place the V9 manifest/test cases, predictive matrix and scripts in the current checkout.
2	Run: python repo_hardened_audit.py --repo . --manifest HARDENED_MASTER_MANIFEST_V9.json --run-gates
3	Resolve every parse, stale-reference, manifest or gate failure. No waivers.
4	Run: python oracle_matrix_runner.py validate; then oracle_matrix_runner.py list
5	Export canonical program JSON per schedule and compare with oracle_matrix_runner.py compare --strict-provenance.
6	Export canonical SSI rows and compare with oracle_matrix_runner.py compare-ssi (source SHA required; use --ordered when order is an oracle field).
7	Store source/oracle hashes, commit, tool versions, AFT/AFW/SSI settings, dates, calendars and tolerances.
8	Promote only readiness-green cases to exact parity.

Required assets and unresolved gates

Severity	Gap	Required action
BLOCKER	Four SSI reports do not have an exact dated source MPP snapshot.	Original/progress UID4 source snapshots and the exact two UID152 dependent-analysis snapshots, or regenerated SSI exports from retained source files with settings captured.
BLOCKER	Current Hard_File_updated4 24-hour binary cannot be semantically compared with archived updated3 24-hour payload.	Download/checkout current MPP and run semantic fingerprint comparison through the same MPXJ version.
BLOCKER	Current repository reference-upload head was not locally executed.	Run repo_hardened_audit.py and the complete quality/parity gate in a current checkout.
HIGH	ref2/2345.afw cannot identify which Project5_TAMPERED binary was loaded.	Acumen export/workspace manifest containing full source path and SHA-256, or retain one canonical source and regenerate the workspace.
HIGH	No XER files exist in the uploaded project-library archives.	Non-CUI P6 XER snapshots plus matching Acumen/SSI/P6 oracle exports and calculation/calendar settings.

Severity	Gap	Required action
HIGH	Large Test File OverAl Fixed report name conflicts with exact schedule-date evidence.	Confirm source provenance or regenerate SSI export with source filename embedded; do not use filename as ground truth.
MEDIUM	Acumen workbooks generally do not embed source SHA-256.	Add a Source Manifest sheet/sidecar with schedule hash, report hash, tool version, metric-library hash, settings, status date, timezone, and export time.
MEDIUM	Stochastic SSI/Acumen SRA cannot be guaranteed bit-exact without vendor RNG details.	Capture seed, trials, distribution, correlation, risk mapping, calendar, focus event, and vendor version; use statistical rather than byte-exact distribution acceptance.

Deliverables

- HARDENED_CORPUS_CROSS_REFERENCE.xlsx
- HARDENED_MASTER_MANIFEST_V9.json
- HARDENED_ASSOCIATIONS_V5.csv
- HARDENED_TEST_CASES_V9.json
- PREDICTIVE_ORACLE_MATRIX.json
- oracle_matrix_runner.py
- repo_hardened_audit.py
- verify_hardened_v9.py
- HARDENED_VERIFICATION_RESULT_V9.json

Traceability

- Repository: polittjd/Schedule-Manipulation-Analysis-Tool-Experiment.
- Current head reviewed: dcacbf4458f2049aeec01be345b32d1685dae27c.
- Last documented green code head: 2dc369678dfc294db189d1bc706eba4ab02b752a.
- Key repository evidence: HANDOFF, REPO-INVENTORY, pyproject, CI workflow, parity tests, formula audit and golden case files.
- Project library: Reference Files.zip and Reference Files 2(1).zip, recursively extracted and independently reverified.